

HUISEOK MOON

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RESEARCH INTERESTS

Assistive and wearable robotics, physical human–robot interaction, multimodal human-motion and intent inference, and nonlinear/model-based/adaptive control. Current research integrates physics-based modeling and simulation with real-time machine learning and multimodal sensing to understand, predict, and assist human motion.

Keywords: Wearable/Assistive Robotics • Machine Learning • Multimodal Sensor Integration • Nonlinear Control

EDUCATION

Université Paris-Est Créteil

Ph.D. in Signal, Image and Control — Advisor: Dr. Samer Mohammed

Thesis: Human-in-the-loop Paradigms for Mobility Assistance Using Powered Exoskeleton

Créteil, France

Nov 2019 – Jan 2024

Sogang University

M.S. in Mechanical Engineering — Advisor: Dr. Kyoungchul Kong

Thesis: Control Algorithm of a Robotic Leg for Reducing Interaction Force Between a Load and the Human Body

Graduate Research Fellowship, Ilwun Science and Technology Foundation (2017–2019)

Seoul, South Korea

Mar 2017 – Feb 2019

Kwangwoon University

B.S. in Robotics (Intelligence System)

Thesis: Research on Personal Mobility Using Ball Balancing Robot

Seoul, South Korea

Mar 2010 – Feb 2017

RESEARCH EXPERIENCE

Postdoctoral Researcher

Université Paris-Est Créteil

Jul 2026 – Sep 2026

Vitry-sur-Seine, France

- Designed a framework for **human movement representation** and **real-time joint moment estimation**, learning shared patterns across walking conditions and using wearable sensor data to estimate ankle moments.
- Developed a **multi-IMU gait assessment system** and conducted experiments involving people with **post-stroke hemiparesis** to evaluate lower-limb kinematics and measurement reliability across walking conditions.

Postdoctoral Researcher

Korea Advanced Institute of Science and Technology (KAIST)

Sep 2025 – Mar 2026

Daejeon, South Korea

- Investigated **physics-informed** human movement analysis and **representation learning** using a multimodal biomechanics dataset.

Research Engineer

Université Paris-Est Créteil

Sep 2023 – Aug 2025

Vitry-sur-Seine, France

- Developed **LSTM-based gait recognition** and adaptive ankle assistance using two foot-mounted IMUs, achieving **98% mean recognition accuracy** across five gait modes during assisted walking in healthy participants.
- Developed **adaptive FES for foot-drop correction** in patients, using **EMG signals** to analyze muscle dynamics and adapting stimulation timing and intensity in real time for energy-efficient assistance.
- Analyzed IMU signals from a portable altimeter to quantify forearm pronation–supination for **objective assessment of parkinsonian motor impairment**.

ATER (Teaching and Research Assistant)

Université Paris-Est Créteil

Nov 2022 – Aug 2023

Vitry-sur-Seine, France

- Led **undergraduate laboratory sessions (TP)** on wearable sensor integration and biomechanical signal processing using Python and MATLAB.

Research Assistant

Université Paris-Est Créteil

Nov 2019 – Oct 2022

Vitry-sur-Seine, France

- Developed nonlinear control strategies for lower-limb robotic assistance, including **impedance modulation** for sit-to-stand movements and **adaptive fuzzy control and model predictive control** for reference tracking in actuated ankle and knee orthoses during walking. Integrated IMU, EMG, and FSR data to adjust assistance based on user states and task demands.
- Combined **online random-forest gait classification** with task-specific ankle torque assistance across level walking, stair negotiation, and ramp traversal.

Research Assistant

Sogang University

Mar 2017 – Feb 2019

Seoul, South Korea

- Developed **full lower-limb exoskeleton control** to reduce interaction forces between the load and the body in high-mobility robotics, analyzed human gait using **OpenSim and Vicon**, and participated in a quadruped robot project.

PUBLICATIONS

Journals

- 2026 X. Luo*, **H. Moon***[†], J.-M. Gracies, W. Huo[†], and S. Mohammed[†]. Knee-Guided Phase–Magnitude Decoupled Adaptive Functional Electrical Stimulation for Drop-Foot Correction. Submitted to *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2026. (Under Review) * Equal contribution; [†] Corresponding author.
- 2026 L. Marusu, C. Gault-Colas, **H. Moon**, S. Mohammed, and J.-M. Gracies. Changes in Walking Speed, Step Length, and Lower-Limb Kinematics over Eight Weeks of Guided Self-Rehabilitation Contracts in Hemiparetic Patients: A Prospective Cohort Study. Submitted to *Journal of Neural Engineering*, 2026. (Under Review)
- 2026 O. Bey, M. Chemachema, R. Jradi, **H. Moon**, H. Rifai, Y. Amirat, and S. Mohammed. Non-Model-Based Finite-Time Adaptive Neural Output Feedback Control for an Active Ankle–Foot Orthosis. *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 14103–14119, 2026. 10.1109/TASE.2026.3718534
- 2025 **H. Moon**, O. Bey, A. Boubezoul, L. Oukhellou, and S. Mohammed. Real-Time LSTM-Driven Dynamic Gait Mode Detection for Enhanced Control of Actuated Ankle–Foot Orthosis. *IEEE Transactions on Robotics*, vol. 41, pp. 4794–4809, 2025. 10.1109/TRO.2025.3593111
- 2023 G. Khodabandelou, **H. Moon**, Y. Amirat, and S. Mohammed. A Fuzzy Convolutional Attention-Based GRU Network for Human Activity Recognition. *Engineering Applications of Artificial Intelligence*, vol. 118, Art. no. 105702, 2023. 10.1016/j.engappai.2022.105702
- 2022 **H. Moon**, R. Maiti, K. Das Sharma, Y. Amirat, P. Siarry, and S. Mohammed. Hybrid Half-Gaussian Selectively Adaptive Fuzzy Control of an Actuated Ankle–Foot Orthosis. *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 9635–9642, 2022. Presented at ICRA 2023 (oral presentation). 10.1109/LRA.2022.3191187
- 2022 I. Jammeli, A. Chemori, **H. Moon**, S. Elloumi, and S. Mohammed. An Assistive Explicit Model Predictive Control Framework for a Knee Rehabilitation Exoskeleton. *IEEE/ASME Transactions on Mechatronics*, vol. 27, no. 5, pp. 3636–3647, 2022. 10.1109/TMECH.2021.3126674
- 2022 W. Huo, **H. Moon**, M. A. Alouane, V. Bonnet, J. Huang, Y. Amirat, R. Vaidyanathan, and S. Mohammed. Impedance Modulation Control of a Lower-Limb Exoskeleton to Assist Sit-to-Stand Movements. *IEEE Transactions on Robotics*, vol. 38, no. 2, pp. 1230–1249, 2022. 10.1109/TRO.2021.3104244

Conference Proceedings

- 2024 O. Bey, R. Jradi, **H. Moon**, H. Rifai, K. Das Sharma, Y. Amirat, and S. Mohammed. A Novel Funnel-Based L_1 Adaptive Fuzzy Approach for the Control of an Actuated Ankle Foot Orthosis. *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 17487–17493, 2024. 10.1109/ICRA57147.2024.10610584
- 2022 **H. Moon**, A. Boubezoul, L. Oukhellou, Y. Amirat, and S. Mohammed. Online Human Intention Detection through Machine-Learning Based Algorithm for the Control of Lower-Limbs Wearable Robot. *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, pp. 809–814, 2022. 10.1109/Humanoids53995.2022.10000150
- 2021 P.-G. Jung, W. Huo, **H. Moon**, Y. Amirat, and S. Mohammed. A Novel Gait Phase Detection Algorithm for Foot Drop Correction through Optimal Hybrid FES–Orthosis Assistance. *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 10391–10397, 2021. 10.1109/ICRA48506.2021.9561497

ACADEMIC SERVICE

Peer Review	IEEE Transactions on Robotics, IROS, ICRA, Engineering Applications of Artificial Intelligence, Mechatronics, and others
Invited Talks	REHAssist Visiting Seminar, EPFL, Switzerland (2025); LISSI–BIOTN Joint Seminar, UPEC, France (2024); LISSI Research Seminars, UPEC, France, 4 sessions (2020–2023)
Volunteering	IFAC World Congress, volunteer (2025); IUT Research Day, poster exhibition (2024); Fête de la Science, public outreach exhibitor (2023); Korea Defense Science & Tech Fair, exhibitor (2017)
Mentoring	M.S. students: L. Marusu (2025), A. A. Benhamlaoui (2024), D. Plet (2024), W. Legendre (2023), J. M. Fadous (2022), R. Ravaka & M. Diaby (2021). Topics spanning multi-sensor gait analysis, AI-based gait event detection, Parkinson’s diagnostics, and assistive robot control

SKILLS

Programming	Python, C, C++, MATLAB, LabVIEW, SolidWorks, LaTeX
Frameworks	PyTorch, TensorFlow, Simulink
Expertise	Real-time ML deployment, Multimodal sensor fusion, Embedded systems, Nonlinear control, Human–robot interaction, Wearable robotics, Signal processing
Languages	Korean (native), English (fluent), French (basic)

REFERENCES

Dr. Samer Mohammed Associate Professor LISSI, Université Paris-Est Créteil <i>Ph.D. Advisor</i> samer.mohammed@u-pec.fr	Dr. Kyoungchul Kong Associate Professor KAIST <i>M.S. Advisor</i> kckong@kaist.ac.kr	Dr. Jaegul Choo Associate Professor Kim Jaechul Graduate School of AI, KAIST <i>Postdoc Advisor</i> jchoo@kaist.ac.kr
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